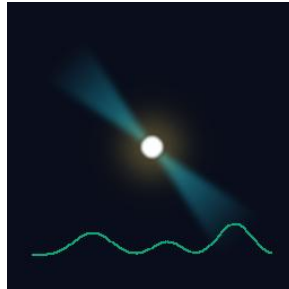


B210 Spectrum Analyzer

Installation Guide



Version 1.0.0

Richard M Hambly (K0GD)

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B210 Spectrum Analyzer — Installation Guide

Version 1.0.0 Author: Richard M Hambly (K0GD) — rick@cncssys.com License: GPL-3.0-or-later

This document covers:

- How to install the B210 Spectrum Analyzer on Windows 11, Linux, and macOS (Intel and Apple Silicon).
- How to update to a new version.
- Common troubleshooting and a reference for advanced settings.

The application itself is one Python file plus a few launchers and an icon. The heavy machinery — GNU Radio, UHD, SoapySDR, PySide6, NumPy, SciPy — is supplied by **Radioconda**, which you install separately as a one-time prerequisite.

Supported radios. The program is designed around the Ettus USRP B210 but also drives any SoapySDR-compatible receiver: SDRPlay RSP1A / RSP1B / RSPduo / RSPdx (with one extra setup step — see §1A), RTL-SDR, HackRF, Airspy / Airspy HF+, BladeRF, LimeSDR, and PlutoSDR. The sample-rate combo and gain slider adapt automatically to whichever device you pick at startup.

1. What the recipient needs to know first

Each recipient performs **two installs**, in this order (plus a third optional step if you're using SDRPlay):

1. **Install Radioconda** (one-time). Radioconda is a curated conda distribution that bundles GNU Radio, UHD, SoapySDR, and the surrounding scientific Python stack for software-defined radio. We require version 4.8 or newer (it ships with UHD ≥ 4.8). Download: <https://github.com/ryanvolz/radioconda/releases>.
2. **Install this app**, which is just a small zip of Python and launcher scripts.
3. **If using SDRPlay only:** install the SDRplay API + SoapySDRPlay module (see §1A). Other supported radios work without any extra setup.

Updates to the app afterwards are a small zip replace — Radioconda does not need to be re-installed for every release.

System requirements

Component	Minimum
OS	Windows 11; Ubuntu/Debian/Fedora/Arch (current); macOS 11 Big Sur or newer
CPU	x86_64 or Apple Silicon (M1/M2/M3/M4)
RAM	4 GB free
USB	One USB 3.0 port (5 Gbps) for the B210
Hardware	Ettus USRP B210 or any SoapySDR-supported receiver (SDRPlay RSPx, RTL-SDR, HackRF, Airspy, BladeRF, LimeSDR, PlutoSDR). With one radio attached the app uses it silently; with several, a

	picker appears. See §8 for details.
Disk	~3 GB for Radioconda, plus ~1 MB for this app
Display	1280 × 800 minimum; 14" MacBook Pro is the layout target

1A. Optional: extra setup for SDRPlay (RSP1A / RSP1B / RSPduo)

If you're using an **SDRPlay** receiver (RSP1A, RSP1B, RSPduo, or RSPdx), you need two extra pieces beyond Radioconda. **Skip this section entirely if you're only using a B210, RTL-SDR, HackRF, Airspy, BladeRF, LimeSDR, or PlutoSDR — those work out of the box once Radioconda is installed.**

1A.1 Install the SDRplay API

SDRPlay devices need the manufacturer's proprietary API driver. Download and install it from <https://www.sdrplay.com/api/> — pick the installer for your OS.

- **Windows:** run the .exe installer; accept defaults. A reboot may be needed.
- **Linux:** run the .run installer with sudo.
- **macOS:** run the .pkg installer.

After installation, verify the API can see your unit with the SDRplay Service / Status app (Windows) or SDRplayService (Linux/macOS). You should see your RSP listed.

1A.2 Install the SoapySDRPlay module

This is the glue between SDRplay's API and the SoapySDR layer the program uses.

Windows / Linux / macOS — from an activated Radioconda prompt:

```
conda install -c conda-forge soapysdr-module-sdrplay
```

Verify it loaded:

```
SoapySDRUtil --info
```

...should list sdrplay in the "Available factories" line. If it doesn't, the API install in §1A.1 didn't take — recheck that.

When you launch the spectrum analyzer with an RSP attached, it will appear in the device picker as e.g. RSPduo – 1234567 [sdrplay].

2. Installing Radioconda

2.1 Windows 11

4. Download radioconda-Windows-x86_64.exe from the Radioconda releases page.
5. Double-click the installer. Accept defaults; the installer offers to put Radioconda under C:\ProgramData\radioconda (system-wide) or %LOCALAPPDATA%\radioconda (per-user). Either works; the app's launcher checks both.

6. After the install finishes, plug the B210 into a USB 3 port. The first time you do this, Windows will silently install the UHD driver from the Radioconda install. No prompt should appear.
7. Verify the B210 is detected by opening the **Anaconda Prompt (Radioconda)** shortcut from the Start menu and running:

```
uhd_find_devices
```

You should see one device with `serial: 3273A91`.

2.2 Linux

8. Download `radioconda-Linux-x86_64.sh` from the releases page.
9. From a terminal:

```
bash radioconda-Linux-x86_64.sh
```

Accept the license and let it install to `~/radioconda` (the launcher checks this path automatically).

10. **USB permissions for the B210.** Radioconda installs the Ettus udev rules, but they need to be activated:

```
sudo cp ~/radioconda/lib/uhd/uhd-usrp.rules /etc/udev/rules.d/  
sudo udevadm control --reload-rules && sudo udevadm trigger
```

Without this step, you'll get permission errors when the B210 is plugged in.

11. Add yourself to the `usrp` (or `plugdev`, depending on distro) group if your distribution uses one. Log out and back in.
12. Plug in the B210 and verify:

```
~/radioconda/bin/uhd_find_devices
```

2.3 macOS (Intel and Apple Silicon)

Radioconda publishes builds for both architectures:

- `radioconda-MacOSX-x86_64.sh` — Intel Macs.
- `radioconda-MacOSX-arm64.sh` — Apple Silicon Macs (M1/M2/M3/M4).

There is **no functional difference** between the two for our purposes. Pick the file that matches your CPU (`uname -m` prints `x86_64` or `arm64`).

13. From a terminal:

```
bash radioconda-MacOSX-<arch>.sh
```

14. Accept the default install location (`~/radioconda`).
15. Plug in the B210. macOS does not require a driver install.

16. Verify:

```
~/radioconda/bin/uhd_find_devices
```

17. **Quarantine note.** If you ever see a Gatekeeper “cannot verify developer” dialog on a Radioconda binary, run:

```
xattr -dr com.apple.quarantine ~/radioconda
```

3. Installing the B210 Spectrum Analyzer

You will have received a zip file named `b210-spectrum-analyzer-<version>.zip` (e.g. `b210-spectrum-analyzer-1.0.0.zip`).

3.1 Windows 11

18. Extract the zip anywhere you have write permission. A common choice is `Documents\B210-Spectrum-Analyzer`. The extracted folder will be `b210-spectrum-analyzer-1.0.0\` and will contain `b210_spectrum_analyzer.py`, `launcher.bat`, `launcher.ps1`, `install-shortcut.ps1`, `LICENSE`, the `icons\` folder, and this guide.
19. Double-click **launcher.bat** to start the application.
20. The first time you run it, the launcher searches for Radioconda in this order: the `RADIOCONDA_ROOT` environment variable, any currently-activated conda env, `%LOCALAPPDATA%\radioconda`, `C:\ProgramData\radioconda`, `%USERPROFILE%\radioconda`, a cached config file, then `conda info -base` if conda is on PATH. If none of these find a working install, you’ll get a prompt asking for the path; type it in and the launcher remembers it for next time.
21. (Optional) Right-click **install-shortcut.ps1** → “Run with PowerShell” to put a “B210 Spectrum Analyzer” shortcut on your desktop and Start menu.

After the first run, the app’s window opens with the spectrum and waterfall plots. Tuning, sample-rate, gain, and recording controls are in the sidebar on the right.

3.2 Linux

22. Extract the zip:

```
unzip b210-spectrum-analyzer-1.0.0.zip  
  
cd b210-spectrum-analyzer-1.0.0  
  
chmod +x launcher.sh
```

23. Launch:

```
./launcher.sh
```

24. (Optional) Install a desktop entry so the app appears in your application menu:

```
INSTALL_DIR="$(pwd)"

sed "s|__INSTALL_DIR__|$INSTALL_DIR|g" b210-spectrum-analyzer.desktop \
    > ~/.local/share/applications/b210-spectrum-analyzer.desktop

update-desktop-database ~/.local/share/applications/ 2>/dev/null || true
```

(The .desktop file uses __INSTALL_DIR__ as a placeholder so the launch path is correct on whichever machine installs it. The sed line substitutes the current directory into the placeholder.)

3.3 macOS

The steps are the same as Linux:

```
unzip b210-spectrum-analyzer-1.0.0.zip
cd b210-spectrum-analyzer-1.0.0
chmod +x launcher.sh
./launcher.sh
```

On Apple Silicon, **make sure** you installed the arm64 build of Radioconda. Mixing an x86_64 Radioconda with a native arm64 Python or Qt will produce confusing errors at startup. There is no Rosetta-only step required if both halves match the CPU.

If the Finder warns about an unidentified developer when running `launcher.sh`, clear the quarantine attribute on the unzipped folder:

```
xattr -dr com.apple.quarantine b210-spectrum-analyzer-1.0.0
```

4. First-run checklist

Regardless of OS, before declaring the install good:

- The window title bar reads B210 Spectrum Analyzer – v1.0.0 – B210 Spectrum Analyzer. The version number must match the bundle you installed.
- Pull up **Help** → **User Guide** from the menu bar. The guide should open.
- Pull up **Help** → **About**. The author, version, license, and the path to the settings file should be readable.
- The spectrum plot should show live data (not a flat line at –140 dB). If it's flat, the B210 isn't streaming — see Troubleshooting §6.
- Click the **408 MHz** preset under Pulsar Band. The center frequency should retune and the trace should redraw.

5. Updating to a new version

Auto-update notifications

The program checks for new versions at launch (no more than once per 24 hours). If a newer version is available, a small dialog opens with the release notes, a button to **Open Download Page** in your browser, a **Skip this version** button (won't nag again about that specific version), and a **Remind me later** button.

You can also trigger a check at any time via **Help → Check for Updates....**

The check is read-only — the program never auto-downloads or auto-installs anything. To disable the auto-check entirely, edit your `settings.ini` and set `auto_check = false` under `[updates]`.

Performing the update

Each release is a self-contained zip. To update:

25. **Close** the running app.
26. **Extract** the new zip alongside the old one (or over the top of the old folder). The new folder is named with the new version (e.g. `b210-spectrum-analyzer-1.0.1`).
27. **Run** `launcher.bat` or `launcher.sh` from the **new** folder.

What is preserved across versions

Your settings and window geometry are kept outside the app folder, so updates never wipe them. The on-disk locations are:

OS	Settings file
Windows	%APPDATA%\B210Analyzer\settings.ini
macOS	~/Library/Application Support/B210Analyzer/settings.ini
Linux	~/.local/share/B210Analyzer/settings.ini

The settings file is plain text and editable with any editor while the app is closed. To reset everything to defaults, either delete the file or use **Help → About → Restore Defaults...** in the running app.

Window geometry is stored separately by Qt

(`HKEY_CURRENT_USER\Software\gnuradio\flowgraphs\b210_spectrum_analyzer` on Windows; the equivalent `QSettings` storage on Mac/Linux). You don't need to touch it.

Backing up your settings

The settings file is small (~1 KB). Copy it somewhere before a major upgrade if you want a quick rollback path:

```
Windows: copy "%APPDATA%\B210Analyzer\settings.ini" "%USERPROFILE%\Desktop\settings.ini.bak"

macOS: cp "~/Library/Application Support/B210Analyzer/settings.ini"
~/Desktop/settings.ini.bak
```

```
Linux:    cp ~/.local/share/B210Analyzer/settings.ini ~/Desktop/settings.ini.bak
```

6. Troubleshooting

“Radioconda not found in standard locations”

The launcher couldn’t auto-detect a Radioconda install. Either:

- Install Radioconda to one of the standard paths the launcher checks (see §3.1 / §3.2 / §3.3), **or**
- Type the path when the launcher prompts you on first run, **or**
- Set the RADIOCONDA_ROOT environment variable to the install root before launching.

“No UHD Devices Found”

The OS doesn’t see the B210. In order:

28. Unplug the B210 from USB. Wait 5 seconds. Plug it back in. UHD’s USB endpoint can hang on some systems after an abrupt kill of the previous run.
29. Make sure you’re using a USB 3 port (not USB 2 — the B210 will be flaky at high sample rates).
30. From a terminal:

```
Windows:  C:\ProgramData\radioconda\Library\bin\uhd_find_devices.exe
Linux:    ~/radioconda/bin/uhd_find_devices
macOS:    ~/radioconda/bin/uhd_find_devices
```

If uhd_find_devices doesn’t see it either, the problem is below our software — check the B210’s LEDs and try a different USB cable.

Persistent “O” overflows at high sample rates

Some “O” characters in the Overflow box on the right sidebar at startup are normal as the USB pipe warms up. A continuous stream during steady-state operation usually means the host can’t keep up with the sample rate:

- USB 2 ports cannot sustain 20 or 25 MS/s. Move to a USB 3 port.
- Other USB devices on the same controller (especially other 5 Gbps devices, or a heavily used external drive) compete for bandwidth. Move the B210 to its own controller if you have one.
- Reduce sample rate to 10 MHz or 16 MHz.

The Overflow box auto-clears after 15 seconds of no new overflows, so once the rate is stable you should see the box empty.

Spectrum looks flat at –140 dB

The GR flow graph is running but no samples are arriving. Usual causes:

- The B210 is in a powered-but-disconnected state. Unplug and replug.

- Antenna is disconnected (this app uses RX2 by default). Confirm a cable is on RX2.
- Sample rate is set to a value the B210 can't actually achieve. Try 20 MHz.

macOS: “developer cannot be verified”

```
xattr -dr com.apple.quarantine /path/to/b210-spectrum-analyzer-1.0.0
```

Settings won't persist

Settings should be saved automatically on every change and on app close. If they aren't:

- Check the path printed in **Help → About**. The file should exist and be writable.
- If the directory's parent doesn't exist, the app will fail to create it silently. On first run after install, click **Help → About → Open Settings Folder** to verify it opens cleanly.

7. Appendix A — Settings file reference

The settings INI is plain text and editable while the app is closed. Sections:

- [tuning] — preset/coarse/fine/manual frequency in Hz.
- [rx] — sample rate in Hz, RX gain in dB, device_driver (e.g. uhd_b200, sdrplay, rtl-sdr), device_serial.
- [recording] — SigMF recording folder.
- [spectrum] — FFT size, window, averaging, max/min hold, Y-axis range, grid, axis-label toggles, dark/light background, trace styling for each background.
- [waterfall] — intensity range, colormap (per background), grid/axis-label toggles, row count.
- [ui] — control-panel visibility.
- [updates] — auto_check (set to false to disable update notifications), plus internal book-keeping fields the program manages on its own.

Missing keys are filled in from built-in defaults on next launch. The file is rewritten on close with a header comment explaining what it is.

To reset every value to its default: use **Help → About → Restore Defaults...**, or just delete the file and launch the app.

8. Appendix B — Device selection and SigMF playback

Choosing among multiple radios

The program enumerates everything supported at launch — Ettus B210s via UHD, plus any SoapySDR-recognised receiver (SDRPlay, RTL-SDR, HackRF, Airspy, Airspy HF+, BladeRF, LimeSDR, PlutoSDR). Behavior:

- **No radio attached, no playback sample:** an error dialog says “No radio found” and explains how to enable playback (see below). The app exits.

- **No radio attached, but `sample.sigmf-data` + `sample.sigmf-meta` are present next to the program:** the app falls back to **SigMF playback mode** (see below). An informational dialog announces the fallback.
- **One radio attached:** the app opens it and remembers it in `settings.ini`.
- **One B210 attached alongside other types** (e.g. B210 + RSPduo, B210 + RTL-SDR): the app silently opens the B210. The B210 is the primary supported device; other types are still reachable through the **Device** button on the sidebar.
- **Multiple B210s attached:** a picker shows the B210s; pick one. Other attached non-B210 radios are reachable through the **Device** button after launch.
- **Multiple non-B210 radios attached (no B210):** a picker shows them all (e.g. an RSPduo and an RTL-SDR); pick one. The app remembers your choice.

If the remembered device isn't attached on a later launch, the picker re-opens.

The **RX** sidebar group has a button labeled `Device: <product> <serial>` showing which radio the current session is using. Clicking it re-opens the picker; the new choice takes effect on the next launch.

The sample-rate combo and gain-slider range adapt automatically to whichever radio is open. SDRPlay caps at 10 MHz; RTL-SDR maxes around 3.2 MHz; HackRF goes to 20 MHz; the B210 to 25 MHz. Saved gain is clamped to the new device's range if you switch to a narrower one.

To pin a specific device permanently without using the picker, edit `settings.ini` and set `device_driver` + `device_serial` under `[rx]`. Use `device_serial = auto` to restore the default "first found" behavior.

SigMF playback mode (no-device fallback)

The distribution bundle includes a short SigMF recording so that users without a B210 can still launch the program, see live spectrum, and try the controls.

- Filenames: `sample.sigmf-data` (raw IQ) and `sample.sigmf-meta` (JSON metadata). Both live next to `b210_spectrum_analyzer.py` in your install folder.
- The file is read in a continuous loop, paced to match the original capture's sample rate.
- The window title shows **[Playback]**, the Device button shows the file name, and the Recording-status label reads `Playback (looping): sample.sigmf-data`.
- **Sample Rate**, **RX Gain**, and **Recording** controls are **disabled** — they have no meaning for a recorded file. Sample rate comes from the file's metadata.
- **Tuning IS enabled** in playback mode and works as a digital frequency shift on the file's baseband. Tuning to the file's actual center frequency (read from the `.sigmf-meta`) shows the recording's true content. Tuning to other frequencies within $\pm(\text{sample_rate}/2)$ of the file's center lets you explore the recorded bandwidth — useful for poking around inside the capture. Tune well outside that window and you'll see noise / wrap-around, since the recording doesn't contain data at those frequencies. This is the expected behavior for true I/Q data, not a bug.

- All visualization controls (Spectrum panel, Waterfall panel, including the dark/light background toggle, FFT averaging, max/min hold, intensity range, colormap, etc.) work normally.
- To replace the bundled sample with your own recording, do a real recording in live mode, then rename the resulting two files to `sample.sigmf-data` and `sample.sigmf-meta` and drop them next to the program. No source-code change is required.
- To skip playback mode and force an error exit when no B210 is attached, delete or rename either of the two `sample.sigmf-*` files.